

Multiclass Learning and Local Regularization

1 Learning Setup

Definition

A multiclass classification problem consists of:

X input space

Y label space

$\mathcal{H} \subseteq Y^X$ hypothesis class

A hypothesis is a function:

$$h : X \rightarrow Y$$

Training data is sampled from a distribution:

$$S = \{(x_1, y_1), \dots, (x_n, y_n)\}$$

$$(x_i, y_i) \sim D$$

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2 Loss and Risk

Definition

0-1 Loss

$$\ell(h(x), y) = \begin{cases} 0 & h(x) = y \\ 1 & h(x) \neq y \end{cases}$$

True Risk

$$L_D(h) = \mathbb{E}_{(x,y) \sim D}[\ell(h(x), y)]$$

Empirical Risk

$$L_S(h) = \frac{1}{n} \sum_{i=1}^n \ell(h(x_i), y_i)$$

3 PAC Learning Goal

The learner receives sample S and outputs:

$$A(S)$$

such that

$$L_D(A(S)) \leq \epsilon$$

with probability at least

$$1 - \delta$$

Binary Classification

Theorem 1. *A binary hypothesis class is PAC learnable iff it has finite VC dimension. ERM achieves near optimal sample complexity.*

ERM rule:

$$\operatorname{argmin}_{h \in \mathcal{H}} L_S(h)$$

Multiclass Learning Problem

For multiclass classification, ERM does not always work.

Some learnable classes require **improper learners**.

4 Local Regularization

Definition

A **local regularizer** is a function

$$\psi : \mathcal{H} \times X \rightarrow \mathbb{R}_{\geq 0}$$

The learner predicts

$$A(S)(x) \in \{h(x) : h \in \operatorname{argmin}_{h \in L_S^{-1}(0)} \psi(h, x)\}$$

This means we select hypotheses with zero training error and choose the one with smallest penalty.

Key Idea

Local regularization allows the learner to combine behaviors of multiple hypotheses depending on the test point.

Open Problem 1

Open Problem 1. *Can all learnable multiclass hypothesis classes be learned using local regularization?*

If yes, can this be done with optimal sample complexity?

Interpretation:

Binary classification:

$$VC \text{ dimension} \Rightarrow ERM$$

Multiclass classification:

$$? \Rightarrow \text{Local Regularization}$$

Candidate Counterexample

Define

$$X = \mathbb{N}$$

$$Y = \{*\} \cup 2^X$$

Set Construction

Choose sets

$$A, B, C \subseteq X$$

such that

$$|A| = |B| = |C| = k$$

$$|A \cap B| = |A \cap C| = |B \cap C| = k/2$$

$$A \cap B \cap C = \emptyset$$

Hypothesis Definition

Each hypothesis satisfies

$$h(x) = * \quad x \notin A \cup B \cup C$$

On overlaps:

$$x \in A \cap B \Rightarrow h(x) \in \{A, B\}$$

$$x \in A \cap C \Rightarrow h(x) \in \{A, C\}$$

$$x \in B \cap C \Rightarrow h(x) \in \{B, C\}$$

There are

$$2^3 = 8$$

possible hypotheses per triple.

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5 Why the Construction is Tricky

Explanation

The hypothesis class has strong symmetry.
Many hypotheses behave identically on training samples.
Thus the learner cannot easily identify the correct label.

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Why ERM Fails

ERM chooses any hypothesis with minimal training error.
But multiple hypotheses can achieve:

$$L_S(h) = 0$$

Example situation:

Training data only reveals part of set A .

For a new test point $x \in A$:

Another hypothesis might predict label B while still fitting all training data.

Thus ERM may choose the wrong label.

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Why Symmetry Causes Problems

Key Idea

The sets A, B, C are symmetric.
Without observing enough structure in the sample, the learner cannot know which label is correct.

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Open Problem 2

Open Problem 2. *Can the class H_Δ be learned by a local regularizer?*

Two possibilities:

$$H_{\Delta}$$

either

is learnable by local regularization

or

is a counterexample

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How Researchers Might Prove the Result

Two strategies exist.

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Strategy 1: Prove Universality

Show that for every learnable class H

there exists

$$\psi(h, x)$$

that induces a PAC learner.

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Strategy 2: Find Counterexample

Prove that for every local regularizer

$$\psi(h, x)$$

there exists a distribution

$$D$$

such that the learner fails.

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One-Page Concise Summary (Submission Section)

Concise Summary

The goal of multiclass learning is to find a learner A that outputs a predictor with low true risk:

$$L_D(A(S)) \leq \epsilon$$

Binary classification is fully characterized by VC dimension, and ERM provides an optimal learning rule.

For multiclass problems, ERM may fail because some learnable classes require improper learners.

Local regularization extends ERM by introducing a penalty

$$\psi(h, x)$$

depending on both the hypothesis and the test point.

The learner predicts

$$A(S)(x) = h(x)$$

where

$$h \in \operatorname{argmin}_{h \in L_S^{-1}(0)} \psi(h, x)$$

The main open question is whether local regularization can learn all multiclass hypothesis classes.

The paper introduces a structured hypothesis class H_Δ based on overlapping sets A, B, C as a potential counterexample.

ERM fails on this class due to symmetry among hypotheses that fit the training data equally well.

The open question is whether local regularization can overcome this symmetry or whether H_Δ proves that a more powerful learning framework is required.